

INDUSTRIAL CROP PRODUCTION I (AGT 116)
Course lecturer: Mr. Enudeme Nduka Martins and Mrs OLAJIDE Comfort
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OBJECTIVES

By the end of this unit, you will be able to:

- Identify different types of industrial crops.
- Identify areas of production of various industrial crops.
- Understand the botany, of important industrial crops.
- Understand the production techniques of industrial crops in Nigeria.
- Understand the production cycle of major industrial crops in Nigeria.

COTTON (*Gossypium spp.*)



FRUIT/SEED FIBRE

Cotton, natural vegetable fibre is of great economic importance as a raw material for cloth. Its widespread use is largely due to the ease with which its fibres are spun in yarns. Cotton strength, absorbency, and capacity to be washed and dyed also make it adaptable to a considerable variety of textile product. Cotton (*Gossypium spp.*)

CLASSIFICATION

Plant Kingdom – Plantae

Division – Magnoliophyta

Class – Magnoliopsida

Sub-Class – Asteridae

Order – Malvales

Family – Malvaceae

Genus – *Gossypium*

Species - *G. Hirsutum* Upland cotton Species (cont.) –

G. Barbadosense- South American cotton (sea Island cotton).

ORIGIN

Cotton cultivation in the world began from India where cotton has been grown for more than 6,000 years since the pre-harappan period. Cotton from the Harappan civilization was exported to Mesopotamia during the 3rd millennium B.C. and cotton was soon known to the Egyptians as well as becoming a prized trading item from Nubia and Moroe.

Cotton has been spun, woven and dyed since prehistoric times. It clothed the people of ancient India with matchless skill and their uses spread to the Mediterranean countries. In the 1st century, Arab traders brought fine muslin and calico to Italy and Spain. The moors introduced the cultivation of cotton into Spain in the 9th century. Fustians and dimities were woven there 14th century in Venice and Milan, at first with a linen warp. Little cotton cloth was imported to England before the 15th cent., although small amounts were obtained chiefly for candlewicks. Arab traders introduced cotton cultivation to the rest of Africans.

Native Americans skillfully spun and wove cotton into fine garments and dyed tapestries. Cotton fabrics found in Peruvian tombs are said to belong to a pre-Inca culture. In color and texture the ancient Peruvian and Mexican textiles resemble those found in Egyptian tombs. In Peru, cotton was the backbone of the development of coastal cultures such as the Moche and Nazca. The Spanish who came to Mexico in the early 1500s found the people growing cotton and wearing clothing made of it. Thus it can be said that the two general centers of origin of the cotton plant are Indo-China and tropical Africa in the old world and central and South American in the new world. Today, cotton remains a major export of the southern United States, and a majority of the world's annual cotton crop is of the long-staple.

GEOGRAPHICAL DISTRIBUTION AND PRODUCTION

Cotton (*Gossypium* Spp.) is found in the tropical and warm temperate zones of all continents. Their natural habitat is the banks of the water courses of seasonal rivers. Many are perennial. There are two main species in cultivation, namely *Gossypium hirsutum* (upland cotton) and *Gossypium barbadense* (Sea Island cotton). Of the world production, about 80% is supplied by *G.*

hirsutum cultivars and about 15% by *G. barbadense* cultivars. India has the largest area under cotton followed by China.

As of 2007, the ten largest producers of cotton in the world are: China, India, United States, Pakistan, Brazil, Uzbekistan, Turkey, Greece, Turkmenistan and Syria. Out of these mentioned producers, the five leading exporters include: United States, Uzbekistan, India, Brazil and Burkina Faso. The largest non producing importers are: Bangladesh, Indonesia, Thailand, Russia, and Taiwan. In United States, the State Texas leads in total production while the State of California has the highest yield per acre in the world.⁴

ECOLOGY

Cotton is primarily grown in dry tropical and subtropical climates at temperatures between 16°C and 25°C. It is a warm climate threatened by health or freezing temperatures (below 5°C or about 25°C), although its resistance varies from species to species. Excessive exposure to dryness or moisture at certain stages of the plant development (lasting 5 to 7 months may be detrimental to cotton quality and yields, and might also kill the plant. It cannot tolerate shade, particularly at the seedling stage. It does not tolerate very heavy rainfall.

Successful cultivation of cotton requires a long frost-free period of 180-200 days plenty of sunshine, and a moderate rainfall, usually from 600 to 1200mm (24 to 48 inches). Cotton can be successfully grown on all soils except the sandy, saline and waterlogged soils. Soils usually need to be fairly heavy, although the level of nutrients does not need to be exceptional. In general, these conditions are met within the seasonally dry tropics and subtropics in the northern and southern hemispheres, but a large proportion of the cotton grown today is cultivated in areas with less rainfall that obtain the water from irrigation. The area of the United States known as the South Plains is the largest contiguous cotton-growing region in the world. It is heavily dependent on irrigation water drawn from the Ogallala Aquifer. Abundant sunshine during the period of boll maturation and harvesting is essential to obtain a good quality produce. The best cotton lands are mixture of clay and sandy loam.

BOTANICAL DESCRIPTION

Cotton belongs to the family Malvaceae or mallow family which also includes hibiscus, okro, cocoa and the familiar swamp mallow.

Gossypium hirsutum- *Gossypium hirsutum*, known as Upland cotton or Mexican cotton, is the most widely planted species of cotton, in the United States constituting some 95% of all cotton production and noble to Central America and possibly Mexico. Worldwide, the figure is about 95% of all production for this species. The fibres range from 10 to 32mm or more in length and are of medium coarseness. The flowers are creamy white when they first open, but they soon turn pink or red. The lint fibres adhere strongly to the seed. The bolls usually contain four or five locks. A lock is a division of the oval called a locule that contains several developing seed.

Gossypium hirsutum is photoperiod insensitive. It includes a number of varieties with varying fibre length and tolerances to a number of growing condition. The longer staple varieties are called "Long Staple Upland" and the shorter length varieties are referred to as "Short Staple Upland".

Gossypium barbadense, Sea Island Cotton, America Pima cottons or Egyptian cottons have extra-long fine fibres, 38 to 51mm long, or even longer in some cases. The lint is readily detached from the seed. The petals are yellow with a pink spot at the base or claw. The bolls usually contain three locks. It is considered to be a shortday photoperiod sensitive plant. It grows as a small bushy tree and yields cotton with usually long, silky fibres. This plant contains the chemical gossypol, which reduces its susceptibility to insect and fungal damage.

In the wild state, cotton is a perennial plant but most of cultivated cottons are annuals. Cultivated cotton is a herbaceous plant which attains a height of 1– 2 meters. The plant has a tap root with secondary roots that branch laterally from primary root. The main stem is erect and much branched.

There are two buds at the base of each cotton leaf petioles. The true auxiliary bud develops into a vegetative branch (monopodial), which only bears leaves and no flower.

The accessory bud generally develops into sympodial or fruiting branch that bears flowers. There is a tendency for the lower branches to be vegetative and the upper fruiting.

The leaves are spirally arranged on the main stem and the vegetative branches. Each leaf consists of a long petiole and a palmate lamina, which is divided into 3– 9 lobes. The leaves and stems are usually covered with fine hairs.

The flowers develop at the node opposite to a leaf in fruiting branches only. Flowers are large and showy and are surrounded by involucres of large, leafy, bracts (epicalyx).

Flowers consist of pistil; the stamens arranged in a tube-like staminal column that envelope the style of the pistil, five petals and five green sepals joined to form a cup-like calyx. The fruit is the enlarged ovary that develops into three to few oil glands while the bolls of *G. barbadense* are dark green and prominently pitted with many oil glands. When the boll is ripe, the capsule cracks or splits along the line or sutures where the carpels meet, and the cotton within expands greatly in a white fluffy mass. The seeds are dark in colour and may be 24–50. The earliest and latest bolls on a plant are usually smaller, with shorter fiber, than the intermediate bolls, late, unopened bolls are called bollea.

The cotton fibre is simply an elongation or outgrowth of an epidermal cell of the seed coat. There are two types of unicellular hairs or outgrowths.

Fuzz: — in which the hairs are circular in cross-section; at first thin walled, but cellulose is then deposited within the hair until the central cavity of lumen is almost obliterated. These hairs are attached firmly to the seed and cannot be spun. There are shorter outgrowths.

The flowers are arranged on alternate sides of the fruiting branch. Additional flowers appear on a branch at about six-day intervals. The corolla consists of five petals that range in colour from white to yellow to purple in different types. The staminal column bears ten or more or less double rows carrying ninety to one hundred stamens. While the pistil consists of three to five carpels. The stigma tips extend up above the anthers. The Asiatic cottons are classified as *G. arboreum* and *G. herbarium*. The fibres are coarse and short, with a length from 13 to 22mm. Cotton developed by Native Americans, also has short fibres. Upland and Asiatic cottons have shorter boll periods than the Sea Island cottons.

LINT: - the longer outgrowth and the deposition of cellulose is reduced, given a lumen or cavity so that the hairs collapse on drying to form a ribbon. The lint is easily detached from the seed

AGRONOMIC PRACTICE

Planting Material: Cotton can be propagated asexually or sexually. Cotton is propagated through direct planting of cotton seed sexually and asexually on the field. Cotton can also be propagated by means of grafting.

Planting: Planting-Seed is often delinted before planting to allow it to pass more freely and germinate more rapidly. Seed is delinted mechanically with delinting saw pin or by treatment with concentrated sulphuric acid and hydrochloric acid. Linters are a mixture of soft or flaccid fibres and fuzz that escaped removal from the seed in ginning. Often the seed is treated after delinting or without delinting, with a fungicide to control seeding diseases.

The seeds should be planted in well-prepared moist soil with high nutrient supplying capacity. Indeed, the cotton plant is particularly weak and its moisture and nutrient uptake is remarkable. Cotton production tends to exhaust the soil, which may require some soil management practices typically by means of physical adjustments, fertilization, and crop rotation (notably with a culture of leguminous plant and one of cereal). Moreover, the root system of the cotton plant is particularly developed and penetrates downward deeply (its depth can sometimes double the height of the surface stem). Accordingly, cotton should be planted in rich seedbeds that are muddy or argillaceous-sandy, where the taproot would grow downward deeply and develop under favourable conditions.

Seeds are planted at 4-6 per hole and spaced to 25 cm apart 90 cm or 75 cm ridges. A seed rate of 15-25 kg per hectare is recommended. Seedling emergence can occur between one week and a month after planting. During this phase (germination, emergence and seedling growth), the plant needs warm temperature and much moisture (7,000 to 9,000 m³ by hectare), which can be supplied by nature or by means of irrigation. Timely sowing of cotton is the single key factor to

influence the cotton yields. Most of the cotton is planted after mid-July, but least yields are obtained from the crop planted in mid-June.

Especially in United States, cotton is increasingly grown as "irrigated" cotton. Although irrigated cotton farming tends to be more expensive than "dry land" cotton (which relies on rainfall), it generally produces high quality lint with greater uniformity and yield potential. Moreover, the maturation period tends to be shorter than for dry land cotton.

Fertilizer needs of the crop can be met by applying recommended rates of 50kg N, 25kg P₂O₅ and 20kg K₂O/ha. Nitrogen is applied at sowing or shortly after germination. P and K fertilizers should be broadcasted and incorporated in the soil just before planting. Nitrogen is the most critical nutrient for cotton production. Most nitrogen fertilizer should be applied in split applications during reproductive growth to reduce nitrogen losses from the soil. Phosphorus and potassium are much less likely to be deficient to cotton.

Weed Control

Pre-emergent herbicides control early season, small seeded annual weeds when applied during planting or shortly thereafter. Band application over the planted ones during planting reduces chemical costs, but cultivation or post emergent herbicides must be applied to control the weeds between the rows. Common pre-emerged herbicides are fluometuron, prometryn and metachlor. Some post emergent herbicides used in cotton fields are applied as basally directed sprays after the cotton plants are 8-18m tall or taller. Some common post emergent herbicides are MSMA/DSMA, fluometuron, oxyfluorfen and lactafen. Early cultivations are often made with a rotary hoe. Later, shallow cultivations with sweep or blade implements cut off the weeds with little injury to the cotton root.

Cotton is highly vulnerable to weed competition especially in the initial stage of growth. Weeds can be controlled manually or with herbicides e.g. 0.8 to 1.0kg a.i./ha of diuron + 0.8kg a.i./ha of nor flurazone applied pre-emergence.

Organic Cotton

Organic cotton is cotton that is grown without insecticide. Worldwide cotton is a pesticide-intensive crop, using approximately 25% of the world's insecticides and 10% of the world's pesticides. According to the World Health Organization (WHO), pesticide poisoning causes 20,000 deaths annually in developing countries, often linked to cotton farming. Organic cotton farming, which prohibits synthetic agrochemicals and fertilizers, instead uses methods like crop rotation and natural pest control to ensure ecological and social sustainability. While more sustainable, the production of organic cotton is generally more expensive than conventional farming.

Cotton Pests:

Cotton insects are the principal cause of yield losses. Estimates indicate that the yield losses due to insect infestations would amount to almost 15% of world annual product. More than 1300 different species of insect pests attack the crop. Among the most common and endogenous species found in cotton fields are

The pink bollworm (*Pectinophora gossypiella*) was first described in 1843 by W.W. Saunders as *Depressaria gossypiella*, from specimens found to be damaging cotton in India in 1842. The pink worm withdraws nutrients from the inside of the cottonseed and may cause serious yield losses. Although the most severe infestations have occurred in Africa and India, the pink bollworm has been recorded in nearly all cotton-producing countries and is a key pest in many of these areas. It also damages the lint and seeds kernels. Infestations may be reduced by the heating of cottonseeds at about 55°C , as well as by other management tactics, including plantation treatment and destroying of the infested crop. The use of early maturing varieties and resistant varieties is recommended.

The Boll Weevils (*Anthonomus grandis*), also known as bollworm, is most common in American cotton plantations, they are a group of moths whose larvae cause much damage to the cotton boll. Infected bolls fall off before they mature. Residues from a previous cotton crop should be pulled up and destroyed or burnt.

They are probably the most destructive insects but the depredations are confined to areas that have an annual rainfall of more than (740 mm). The bollworm and the pink bollworm also destroy the contents of the boll. Genetically, engineered cotton varieties that contain genes from *Bacillus Thuringiensis* (Bt cotton) are resistant to the tobacco budworm and the pink bollworm and show some resistance to the bollworm.

The Egyptian (spiny) bollworm (*Earias insulana*) and the red bollworm *Diparopsis castanea* feed on the developing cotton bolls.

Cotton stainers (*Dysdercus supersticiosus*) attack maturing cotton bolls and seeds. They may cause the staining of the lint. In addition, feeding wounds may allow the entry to the boll of saprophytic fungi (organisms which draw nutrients from the host, but do not harm it, contrary to parasites). Other insect pests of cotton, such as the white flies (*Bemisia gossypiella*) may adversely affect lint quality and yield potential. They suck sap from leaves and pose the most serious threat in India and Africa.

The cotton aphid (*Aphis gossypii*), also known as the melon aphid, infests the cotton seedlings. Cotton aphids are among the most injuring insects found in cotton. They suck sap from leaves and secrete honeydew on the undersides of leaves. Honeydew secretions may burn the leaves and interfere with photosynthesis. In addition, aphid is a vector of viruses and a carrier of other insects. In Africa, aphid infestations are among the most injuring insect pests in terms of economic yield loss.

DISEASES

Nematodes: There are approximately 128 species of nematodes associated with cotton. Five parasitic forms pose the most serious threat to the crop, including the *Meloidogyne incognita* (or root knot nematode) and the *Rotylenchus reniformis* (or reniform nematode). These two species can become serious pests (in the United States, particularly in the State of Virginia, they accounted for 99% of the damage by cotton parasitic nematodes). These parasites live in the soil (the root knot nematode favours rough and arenaceous soil) and withdraw nutrients from the plant roots. Symptom patterns associated with nematodes include stunting, potassic deficiency or early maturity. Nematodes can reduce yields (in Alabama, United States, yield losses are estimated to average 10% or 20% but can peak to 50% in arenaceous dry soil). Also, depending upon the stage of development of the infested crop, they can hamper the quality of cotton.

HARVESTING

Cotton plantation requires vast labour force to hand pick cotton fibres and it was not until the 1950s that reliable harvesting machinery was introduced. Cotton harvesting before now had been tedious and clumsy and required large labour and man work hour.

Flowering generally starts one month and a half to two months after the crop is planted. Blooming will continue regularly for several weeks, even months, as long as growing conditions are suitable. After flowering, the inner part of the bloom gradually develops into a fruit (called "cotton boll"). Cotton bolls keep growing until full size (approximately 2 to 3cm). It will take about two months between the blooming of the flower and the first opening of the bolls.

Cotton bolls burst open at maturity, revealing soft masses of fibres. Cotton harvesting is then possible. The cotton is picked either manually or mechanically. Manual picking is a very labour intensive and time-consuming task, and may be rather expensive. However, it generally produces quality lint with limited amount of trash, since cotton bolls are picked by hand as they burst open upon maturity.

Cotton is harvested mechanically by cotton pickers (the most commonly used) or cotton strippers, which remove all the cotton bolls.

Cotton strippers are generally used after application of a defoliant. A cotton combine harvest is a picker or scrapper that cuts off the plants, strips out the seed cotton with spindles, and then chops the stalks and branches and returns them to the soil.

Mechanical harvesting is faster than the manual picking of cotton. However, unwanted leaves and twigs may be collected with the cotton. Cotton picked by a stripper might thus need additional cleaning (sorting of the trash) in order to obtain quality lint.

POST HARVESTING HANDLING

During harvesting of cotton with a cotton stripper, unwanted twigs and leaves may be collected with the cotton thus needing additional cleaning (sorting of the trash) in order to obtain quality lint. Once the cotton is picked either manually or mechanically, it is transported to a cotton gin where the cotton fibre (lint) are separated from the cotton seeds. The cotton lint is then compacted in bales and stored.

PROCESSING

Seed cotton is first dried. It is then put into a machine, which separates the lint from the seed and fuzz. Cotton ginning involves separation of the fibres from the seed after removal of dirt, hulls and other trash; damp or wet cotton is put through a drier before ginning. Dry lint contains eight per cent moisture or less. The mechanism that separates the seed from the lint is called a gin stand. At the press or baler, the fluffy lint is pressed into bales of about 230kg. The lint derived after ginning is compressed and packed into bales. The seed, which is left after ginning has the fuzz attached. This is removed mechanically and the clean seed is further processed to yield cotton seed oil. Cotton seed contains 30% starch, 25% semi-drying oil and 16–20% protein.

UTILIZATION/ECONOMIC IMPORTANCE

Gossypol is a natural toxin that helps protect cotton plants from insect damage. It is secreted by glands on the leaves and in the seeds. Cotton seed meal is a high-protein feed, but the presence of gossypol makes it toxic to livestock if fed in large quantities. This limits the proportion of cottonseed meal that can safely be added to livestock rations. Cotton with few glands on the leaves and in the seeds is low in gossypol. Use of glandless cotton, however, reduces the cost of oil processing increasing the protein (meal) quality of the seed meal. About 73 per cent of domestic cotton is used for clothing and household goods.

Cotton lint is spun into thread to be woven into various fabrics. Cotton is used to make a number of textile products. These include:

Terrycloth used to make highly absorbent bath towels and robes.

Denim, used to make blue jeans.

Chambray, popularly used in the manufacture of blue work shirts (from which we get the term "blue-collar"); and corduroy, seersucker, and cotton twill.

Socks, underwear, and most T-shirts are made from cotton. Bed sheets often are made from cotton

Cotton also is used to make yarn used in crochet and knitting.

Fabric also can be made from recycled or recovered cotton that otherwise would be thrown away during the spinning, weaving or cutting process. While many fabrics are made completely of

cotton, some materials blend cotton with other fibres, including rayon and synthetic fibres such as polyester.

In addition to the textile industry, cotton is used in fishnets, coffee filters, tents, gunpowder (see Nitrocellulose), cotton paper, and in bookbinding. The first Chinese paper was made of cotton fiber. Fire hoses were once made of cotton.

The cottonseed which remains after the cotton is ginned is used to produce cottonseed oil, which, after refining, can be consumed by humans like any other vegetable oil. Refined cotton seed oil is used for edible purposes for cooking, salad and manufacture of margarines.

The cottonseed meal or cake that is left generally is fed to livestock. Cotton seed hulls are used as roughage for livestock and as livestock bedding, fertilizer and fuel.

Cotton linters are fine, silky fibres which adhere to the seeds of the cotton plants after ginning. These curly fibres typically are less than 1/8 in, 3mm, long. The term also may apply to the longer textile fibre staple lint as well as the shorter fuzzy fibres from some upland species. Linters are traditionally used in the manufacture of paper and as a raw material in the manufacture of cellulose.

SISAL HEMP (*Agave sisalana*)



CLASSIFICATION

Plant kingdom – Plantae

Division – Magnoliophyta

Class – Magnoliopsida

Sub-class – Liliidae

Family – Agavaceae

Genus – Agave

Species – Sisalana

ORIGIN AND GEOGRAPHICAL DISTRIBUTION

In the 19th century, sisal cultivation (the plant being propagated via offsets), was spread worldwide, from Florida to the Caribbean Islands and Brazil; as well as to countries in Africa, notably Tanzania, Kenya, and Asia. Among, flax, hemp[, abaca, sunn hemp *Crotalaria juncea*) and other agro-based fibre species, annual sisal production is the third largest worldwide, after cotton.

Today, Brazil is the major world producer of sisal at 125,000 tons. Sisal occupies 6th place among fibre plants, representing 2% of the world's production of plant fibres (plant fibres provide 65% of the world's fibres).

Sisal is a native of the Yucatan peninsula, Mexico but found in part of Central America and distributed to the West Indies, where it is being increasingly cultivated. It was introduced into Tanzania in 1893 and into Kenya in 1913

ECOLOGY

Sisal grows best in a hot climatic environment and may be grown throughout the humid and sub-humid lowland tropics. Sisal is essentially a plant of the tropics and sub-tropics and production benefits from temperature about 25 degree Centigrade and plenty of sunshine. Sisal prefers a rainfall of not less than 1000-1500mm. The more even, the rainfall distribution, the higher the quality of fibre and opportunity for continuous leaf harvesting. Excessive rainfall is harmful and the plants cannot tolerate water-logging. The crop generally prefers a medium or light soil with a pH of 5.5.-7.5.

BOTANICAL DESCRIPTION

Sisal is a perennial with a short thick stem or bole that is succulent which with good growing conditions forms an inflorescence after 6-9 years after having produced 200-250 leaves, and then dies. Leaves average 120cm in length and are arranged spirally around the thick stem. The leaves are 75% schlerenchyma bundles. They are thick, fleshy and often covered with a wax bloom. The tip tapers into a sharp purplish black spine. The root system is shallow but extends up to 3.5m from the stem. As a cactus, Agave plants survive and produce a marketable product in infertile arid regions which in many cases would otherwise be unproductive. As it is a labour intensive crop it offers at least stability for a large rural population.

The flowers are borne in dense clusters at the end of short lateral branches and closely resemble those of Agave Americana. After they have begun to wither, buds are developed from the point of union of the flower stalk, these from the tiny plants which when several inches long become detached and fall into the ground.

The sisal plant has 7-10 years life span and typically produces 200-250 commercially usable leaves. Each leaf contains an average of 1000 fibres. The fibre element, which accounts for only about 4% of the plant by weight, is extracted by a process called decortications. After production of the leaves, it sends up a flowering arms or pole nearly 9m tall, branching in the top 1.5m and bearing a dense cluster of flowers. Bulbils develop from tiny buds present on each stalk. Each bubil is a plantlet consisting of a meristem. A pole may produce as many as 3000 bulbils produced from the suckers, from buds in the axile of leaves, from rhizomes which are below ground level.

CULTURAL PRACTICES

Propagation of sisal is by means of bulbils which appear on the flower stalk, or by suckers growing around the base of the plant. Plants produce up to 4000 bulbils/plant compared with less than a quarter of this number of suckers; hence bulbils are preferred for propagation.

Only large bulbils should be selected for planting. Bulbils are first planted into nursery beds at spacing of 25-30cm apart in rows 50cm apart. The bulbils are allowed to grow to about 40cm or until they are about 9-12 months old, after which they are transplanted to the field. At this time the bulbils have good roots. The growth of bulbils is improved by mulching sisal nurseries with grass, paper or polythene. Mulching with partially rotted sisal gives best results.

The crop should be kept weed free and annual legumes such as beans may be cultivated between rows to suppress weed growth and limit erosion. Transplanting takes place preferably at the beginning of the rainy season. A recommended pattern in the field is a series of double rows 60cm apart with a 2.5cm alley between pairs of rows. Plant spacing is at 75cm giving a population of about 25000 plants per hectare.

Alternatively, plant can be spaced 1m apart in 3m rows. Fertilizer application is by the use of super phosphate fertilizer applied at about 25kg per hectare per annum with about 50kg of Potassium; Nitrogen can be applied as calcium ammonium nitrate or urea to ensure a good crop. Apply lime where the pH of the soil falls below 6.5, where rainfall levels are good nitrogen can be supplied by growing leguminous cash crop in-between crops.

It has also been known to have a sizeable demand for calcium as a nutrient and frequently responded to small dressings of lime when grown on acid soils. The crop is also susceptible to boron deficiency. The yield deteriorates over the years and more rapidly over continuous cultivation. Soil potassium deficiencies are indicated by a bending-over of the normally straight and still older leaves.

Boron deficiency which causes leaf cracking is more difficult to control. Decertification wastes including both liquid and solids have proved effective as fertilizer supplement, especially when supplemented with lime. Bulbils are first raised in the nurseries for about 12-18 months before they are planted out in the field. It is important to weed during the first 2-3 years by hand hoeing, mechanically or with herbicides. Atrazine can be applied as a spray after planting. Generally, no commercial fertilizer is applied.

PEST AND DISEASES

The plant has a leathery epidermis and some species have sharp spines for protection. Only pest known to occasionally reach pest proportions is the Sisal Weevil (*Scyphophorus Interstitialis*). The weevil produces two kinds of damage: (1) Puncture feeding on soft tissues of the leaves by the adult and (2) Feeding by the larvae on the young unfurled leaves in the growing spike. The latter is more serious. Aldrin and Dieldrin insecticides that are persistent in the soil

have generally given adequate control, whether applied as spray or dust. However, the weevil is not widespread and only the younger plants appear to be seriously damaged.

Diseases: Sisal has fewer problems with diseases than most crops do. Disease symptoms are often caused by nutrients deficiencies.

In higher rainfall areas particularly, plants are subject to leaf spots. Other diseases are bole rot and zebra disease, but neither has been reported as epidemic.

HARVESTING

In the countries of production an optimal production cutting of the leaves takes place 18-24 months after planting. More usually, plants are harvested after 24-36 months. In the countries of production about 50 leaves, each weighing up to 1 kg may be cut per plant per year. The ripest lower leaves are cut first and this continues periodically over the next four years. On average, over the first 4 years, two cutting are made annually. In following years only one cut is made per year, until the flower stalks begin to develop. A total of about 300 leaves may be harvested during the economic life of each plant, giving a total of 500-600 tonnes fibre/ha. Over the usual 8 year production period, the average annual yield is 67 leaves; about 2.25 tonnes fibre/ha. The crop dies after producing the inflorescence and bearing a crop of vegetative bulbs, when the plant is 8-10 years old.

The cut leaves are then tied in bundles and are transported to the factory centre for decortications. Decortications should be completed within 24 hours after cutting to avoid deterioration of the fibre.

POST HARVEST HANDLING

Fiber Extraction: A process of decortications is used to extract the fibre from the leaf tissues. Leaves are crushed and beaten by a rotating wheel set with blunt knives, so that only fibres remain. All other parts of the leaf are washed away by water. Decorticated fibres are washed before drying in the sun or by hot air. Proper drying is important as fibre quality depends largely on moisture content. Artificial drying has been found to result in generally better grades of fibre than sun drying. Dry fibres are machine combed and sorted into various grades, largely on the basis of the previous in-field separation of leaves into size groups.

In collecting the fibre the leaves are cut off at the base, the spine at the top end removed, and the leaves carried in bundles to the machines. Here two scraping wheels remove the pulp from the leaves. The leaves are put into the machines at one side, and delivered clean at the other. One half is cleaned by the first wheel, then the cleaned portion is held while the second wheel cleans the remainder of the leaf; all the operations are automatically performed.

UTILIZATION AND ECONOMIC IMPORTANCE

Sisal fibres are typed by properties relational to the performance of the fibre. Researcher Sara Kadolph has found that sisal fibres are smooth, straight and yellow and can be long or short. Since sisal is fairly coarse and inflexible, Kadolph finds that sisal is used by itself or in blends with wool and acrylic for a softer hand.

Sisal is valued for cordage use because of its strength, durability, ability to stretch, affinity for certain dyestuffs, and resistance to deterioration in saltwater.

Sisal is used by industry in three grades. The lower grade fibre is processed by the paper industry because of its high content of cellulose and hemicelluloses. The medium grade fibre is used in the cordage industry for making: ropes, baler and binders twine. Ropes and twines are widely employed for marine, agricultural and general industrial use. The higher-grade fibre after treatment is converted into yarns and used by the carpet industry.

Products made from sisal are being developed rapidly, such as furniture and wall tiles made of resonated sisal. A recent development expanded the range even to car parts for cabin interiors. Other products developed from sisal fibre include spa products, cat scratching posts, lumber support belts, rugs, slippers, cloths and disc buffers. Sisal wall covering meets the abrasion and tearing resistance standards of the American Society for Testing and Materials and of the National Fire Protection Association.

Traditionally, sisal has been the leading material for agricultural twine ("binder" and "baler" twine) but the importance of this is diminishing with competition from polypropylene and other techniques evolving.

Apart from ropes, twines and general cordage sisal is used in low-cost and specialty paper, dartboards, buffing cloth, filters, geotextiles, mattresses, carpets, handicrafts, wire rope cores and macramé.

In recent years sisal has been utilized as a strengthening agent to replace asbestos and fiberglass as well as an environmentally friendly component in the automobile industry. Products made from sisal fibre are purchased throughout the world and for use by the military, universities, churches and hospitals. Sisal does not build up static nor does it trap dust, so vacuuming is the only maintenance required. High spill areas should be treated with a fibre sealer and for spot removal, a dry cleaning powder is recommended. Depending on climatic conditions, sisal will absorb air humidity or release it causing expansion or contraction. Sisal is not recommended for areas that receive wet spills, or rain or snow.

KENAF (*Hibiscus cannabinus* L.)

CLASSIFICATION

Plant Kingdom – Plantae

Division – Magnoliophyta

Class – Magnoliopsida

Subclass – Asteridae

Order – Malvales

Family – Malvaceae

Genus – *Hibiscus*

Species – *Cannabinus*

16. ORIGIN AND GEOGRAPHICAL DISTRIBUTION

Kenaf is believed to have originated from Africa (Western Sudan). Grown thousands of years in Africa where its leaves are consumed in human and animal diets, the bast fibre used for cordage, and the woody core of the stalks burnt for fuel; this crop was introduced into Southern Asia around 1900. It is grown for its soft fibres.

Kenaf (*Hibiscus cannabinus*) is a species of *Hibiscus*, probably native to Southern Asia, though its exact natural origin is unknown. The name also applies to the fibre obtained from this plant. Kenaf is one of the allied fibres of jute and shows similar characteristics. Other names include Bimli, Ambry, Ambian hemp, Deccan Hemp, and Bimlipatum jute.

17. Common Names

English: Kenaf (Persian origin)

India (Bengal): mesta

India (Madras): palungi

India (Bombay): deccan hemp

India (Andhra Pradesh): Bimli jute

Taiwan: ambari

Egypt and Northern Africa: til, teal, or teal

Indonesia: Java jute

Brazil: papoula de Sao Francisco

South Africa: stokroos

West Africa: dah, gambo, and rama.

Kenaf has a long history of cultivation for its fibre in India, Bangladesh, Thailand, parts of Africa, and to a small extent in Southeast Europe. The stems produce two types of fibre, a coarser fibre in the outer layer (bast), and finer fibre in the core. It matures in 100 to 1000 days. About 9000 cultivars are produced today, principal farming areas are throughout China, India and in many other countries including the following: Mackay, Australia in trial stages; seed farms Texas, USA and Tamaulipas, Mexico; North Carolina, Senegal to name a few.

Principal production areas are China, India and the Tashkent area of the former USSR. Much of the Kenaf production around the world is still grown and processed under labour intensive practices. Research and development efforts primarily in the USA since 1980 have resulted in a completely mechanized approach to Kenaf, which has reduced labour requirements and environmental impact. However, the principal change required to effect the commercialization of Kenaf is the need to focus on product applications and marketing of products using Kenaf fibres.

The systems approach postulated by Kenaf International, Limited calls for putting more resources into work focusing on market development instead of the standard production research focus. Essentially, Kenaf is a traditional, third world crop that is poised to be introduced as a new annually renewable source of industrial fibre in the so-called developed economies.

Kenaf has a relatively wide range of adaptation to climate and soils. With the exception of some early types developed for the Asiatic regions of the former USSR, most of the current Kenaf varieties and technologies favour growing at low elevation between 37° N and S latitudes

This is a comprehensive transcription of the text provided in the images, organized by the original section headings for clarity. Optimum growth is generally found in areas like the low Rio Grande Valley of South Texas with its long, warm growing season and moderate rainfall backed up by irrigation.

A member of the hibiscus family (*Hibiscus cannabinus* L.) is related to cotton and okra, and grows well in many parts of the U.S. It offers a way to make paper without cutting trees. The plant is photoperiod sensitive; flowering when days are short and growing more vegetative when days are long. It thrives best on well-drained, neutral, sandy loam soils, rich in humus. It does not tolerate water-logging.

19. BOTANICAL DESCRIPTION(morphology, anatomy)

According to Dempsey (1975), Kenaf is a short-day, annual herbaceous plant cultivated for the soft bast fibre in its stem. More recent efforts have identified commercial products from the woody core fibres as well. Kenaf belongs to the Malvaceae, a family notable for both its economic and horticultural importance. The genus *Hibiscus* is widespread, comprising some 200 annual and perennial species. Kenaf is closely related to cotton, okra, and the hollyhocks. Kenaf, along with roselle, is classified taxonomically in the *Furcaria* section of *Hibiscus*. This section includes between 40 and 50 species (distributed throughout the tropics) that are closely related morphologically. The cultivated forms are erect 2.5–4m tall, with well developed taproots. The straight stems can be green, red or purple with alternate leaves which vary in shape from cordate, palmate to oblong lanceolate lobes. The flowers are borne singly in axils of upper leaves and the fruit is a capsule. Seeds are numerous and usually brown.

- * While the flowering can last 3 to 4 weeks, or more, per plant, each individual flower blooms for only one day. The stalk of the Kenaf plant consists of two distinct fibre types.

- * The outer fibre is called "bast" and comprises roughly 40% of the stalk's dry weight. The refined bast fibres measure 2.6mm and are similar to the best softwood fibres used to make paper.

- * The whiter, inner fibre is called "core" and comprises 60% of the stalk's dry weight. These refined fibres measure 6mm and are comparable to hardwood tree fibres, which are used in a widening range of paper products.

20. AGRONOMIC PRACTICES

Pesticides and fertilizer use in Kenaf crops is minimal because Kenaf is considered to be a hardy plant that requires a minimum of fertilizers, pesticides and water in comparison to conventional row crops. Chemical fertilizers and pesticides used in large scale farming cause run-off pollution in rivers, lakes, estuaries, oceans and underground water.

Kenaf is directly seeded with conventional grain drills after soil surface temperature exceeds 55°F and all danger of frost has passed. The germination rate of the seed should be at least 75 to 80%. Seeds are drilled two rows per bed on 40-inch center with intra-row spacing of approximately 2 inches. Effort should be made when introducing a new crop like Kenaf to avoid much change to standard local spacing.

Obviously, the planting configurations must be coordinated with harvesting plans. Land preparation is similar to other agronomic crops, e.g., cotton. Fertilizer and irrigation should be applied as needed. Generally, areas that are known to be heavily infested with root-knot nematodes should be avoided as most of the currently available varieties are susceptible. Moisture requirements vary but are believed to be between 80mm to 160mm per month during the crop's first 100 days.

A well prepared seedbed for Kenaf as with other crops is desirable. Kenaf can be broadcasted or planted in drill row, seeding rate vary from 15kg to 25kg per hectare. Recommendation for row spacing varies from 20cm to 45cm for production of spinnable fibre. Row spacing as wide as 90cm have been used for paper production trials, but firm recommendations have not yet been established.

On a moist well prepared seedbed, kenaf germinates quickly and will shade or crowd out most weeds. However land that is known to be badly contaminated with weed should not be planted with kenaf as this will reduce yield. Fertilizers can be applied at the rate of 300kg/ha of NPK (25-10-0) just before or after sowing.

21. PEST AND DISEASES

The most serious diseases attacking Kenaf are largely fungal in nature, most of which are seed borne or soil borne. Control measures for the most part may consists of initial selection of disease resistant varieties, seed treatment with organic or mercurial fungicides and early harvest of affected areas.

The most serious disease is caused by *Colletotrichum hibisci*, the disease is frequently called anthracnose or leaf blight. Fortunately, most of the varieties in use today were selected for resistance to the then existing strains of *Colletotrichum hibisci*. However, a new and virulent strain of the organism was recently reported in Kenya where some of the normally resistant strains were attacked.

The most serious pest is the root knot nematode *Meloidogyne incognita acrita*. There are no varieties resistant to these nematodes and the damage to the crop can be serious. Planting on new land or on land that has had a crop not attacked by root knot nematode appears to be the best method of control.

Kenaf has several major insects pests among which are the small black flea beetle of the family Crysomelidae found in Southeast Asia, New Guinea and Africa. It attacked leaves and stems of mature Kenaf and is controlled by DDT which also controls a secondary pest the leaf hopper of the family Cleadellidae.

21.1 The European corn borer

Pyranata nubilalis has heavily damaged kenaf planting in Taiwan. Control has been effected by planting corn around the kenaf after the adult lays egg on the capsule. The adult may be controlled with weekly spraying of dieldrion equivalent to 0.5kg of 100% active ingredients per hectare as required. In Iran, the major pest attacking seed capsules is the larva of the spiny bollworm, *Eatias insulana*, which may be controlled by spraying two or three times at weekly intervals with sevin at the rate of 2kg/ha. *Aphis gossypii*, the cotton aphid sometimes attacks the young kenaf and may be controlled by spraying with Malathion at the rate of 2kg per hectare. Kenaf for fibres will not support an extensive spray program for either diseases or pest but spraying is more likely to be advisable for a seed crop which is usually on a much smaller acreage than a crop for fibre.

22. HARVESTING AND POST HARVEST HANDLING

Kenaf International Limited with support from USDA's CSREES agency contracted with H. Willett & Associates, Inc. to design and test a harvesting, handling, hauling, and storage system for large-scale kenaf operations. Having experimented unsuccessfully in the past with various types of forage choppers, the team decided to modify the cane harvesting system used in southern Louisiana. The system was tested in 1988 and has been used in subsequent harvests.

Essentially, the kenaf crop is topped at 3.6cm and laid down to dry in windrows. The semi-dry stalks are picked up about 10 days later, cut into 30cm billets, and blown into an accompanying dump buggy. The full buggies are unloaded into field-side module builders and the modules are then hauled to the fibre yard for storage.

The crop is harvested 3–5 months after sowing. The least time to harvest is when about 10 flowers are in bloom, at which time the fibre is at its best quality and more easily separable.

23. PROCESSING

Kenaf fibres can be processed traditionally using retting ponds as is common in Asia and parts of Africa and Latin America. In the USA the fibres are separated using a process developed by Willett at operations located in Louisiana and Texas.

A third operation has been set up in Mississippi. The objective is to effect a mechanical separation of bast and core fibre materials for a wide range of product applications. The degree of separation required depends upon the end use.

As indicated earlier, the bast fibre can be pulped as a vegetable fibre for specially pulp and paper uses. The bast fibre can also be blended with plastic for injection molding purposes using the technology developed and patented by Fiber Packaging International Inc. Mats can be made from the bast fibre using several textile and fiberglass technologies.

Applications for the mats range from seeded grass mats for instant lawns to moldable mats for manufactured parts and containers. Some researchers are blending bast fibers with cotton for some textile purpose, but it is not clear whether this will be economically viable. Other potential uses include using the washed fibres in handicrafts.

The core fibres currently are sized and sold into several markets including oil absorbent (based on patent issued to H. and C. Willett), soil-less potting mixes, animal bedding, packing materials, organic filler for plastics, additive for drilling muds, and insulation. It is possible to compress the core in board and it might have some specially pulp uses. Some of the core material might be collected after its primary use, say as absorbent or animal bedding and prepared for other uses.

Kenaf fibres can be pulped either in whole stalk or separated forms. Although various pulping technologies have been tested with kenaf, the only commercial operations presently using kenaf are draft (sulfate) mills in Thailand using whole stalk and in North Carolina using bast fibre. Kenaf is not the principal fibre source for either mill. Most of the work to date on Kenaf for newsprint and other printing and writing grades have focused on a modified thermo-mechanical process, which is energy intensive.

24. UTILIZATION AND ECONOMIC IMPORTANCE

The main uses of Kenaf fibre have been rope, twine, coarse cloth (similar to that made from jute) and paper. In California, Texas and Louisiana, 3,200 acres (13km²) of Kenaf were grown in 1992, most of which was used for animal bedding and feed.

Emerging uses of Kenaf fibre include:

- * Engineered wood, insulation and clothing-grade cloth.
- * Panasonic has set up a plant in Malaysia to manufacture kenaf fibre boards and export them to Japan, oil absorbent (based on patent issued to H. and C. Willett), soil-less potting mixes, animal bedding, packing materials, organic filler for blending with plastics for injection molding.
- * As an additive for drilling muds, and various types of mats, such as seeded grass mats for instant lawns to moldable mats for manufactured parts and containers.
- * Kenaf seeds yield a vegetable oil that is edible and high in omega antioxidants.

* The Kenaf oil is also used for cosmetics, industrial lubricants and as bio-fuel.

* The use of Kenaf in paper production offers various environmental advantages over producing paper from trees.

* For pulp the plants should be harvested when they cease to grow or killed by frost. The bast or outer bark fibre and the inner thick core of the plant are extracted for use.

JUTE PLANT (*Corchorus spp*)

1.0 Identification of the Industrial Crop (Jute)

Jute is an important natural fibre crop adapted to humid regions of Nigeria. It requires fertile, well-drained soil, adequate rainfall, proper crop management, and careful harvesting and retting to produce high-quality fibre. With good agronomic practices, it offers good yields and has strong potential as a profitable industrial crop.

Jute is an important industrial fibre crop grown mainly for its strong, natural fibres.

These fibres are used in making:

- Sacks and bags
- Ropes and twines
- Carpets
- Mats
- Packaging materials
- Geotextiles
- Paper

The two major cultivated species are:

- *Corchorus capsularis* (White jute)
- *Corchorus olitorius* (Tossa jute)

1.2 Origin and History of Jute

Jute originated in the tropical regions of South Asia, especially India and Bangladesh.

It has been cultivated for thousands of years. It became an important export crop during the 18th and 19th centuries. Today, India and Bangladesh are the world's largest producers.

In Nigeria, jute is grown on a small scale and has potential for expansion because of its suitable climate.

1.3 Adaptation to Nigerian Climatic Conditions

Jute grows well under these conditions:

- Temperature: 24–35°C

- Rainfall: 1,500–2,500 mm annually
- Relative humidity: 70–90%
- Deep, fertile loamy or alluvial soils
- Soil pH: 6.0–7.5
- Full sunlight
- Good drainage

Suitable areas in Nigeria include:

- Southern Nigeria
- Middle Belt
- River valleys
- Swampy areas with adequate rainfall

2.0 Areas of Production

Main Producing Areas

- Cross River
- Akwa Ibom
- Rivers
- Bayelsa
- Delta
- Edo
- Ogun
- Ondo

Marginal Producing Areas

- Benue
- Kogi

- Niger
- Taraba
- Kaduna (irrigated areas)

2.1 Production Trends in Nigeria

Production is still limited.

Most fibre products are imported.

Farmers mainly grow jute leaves as vegetables.

There is increasing interest in industrial fibre production because jute is environmentally friendly.

Government and private investors are encouraging fibre crop production.

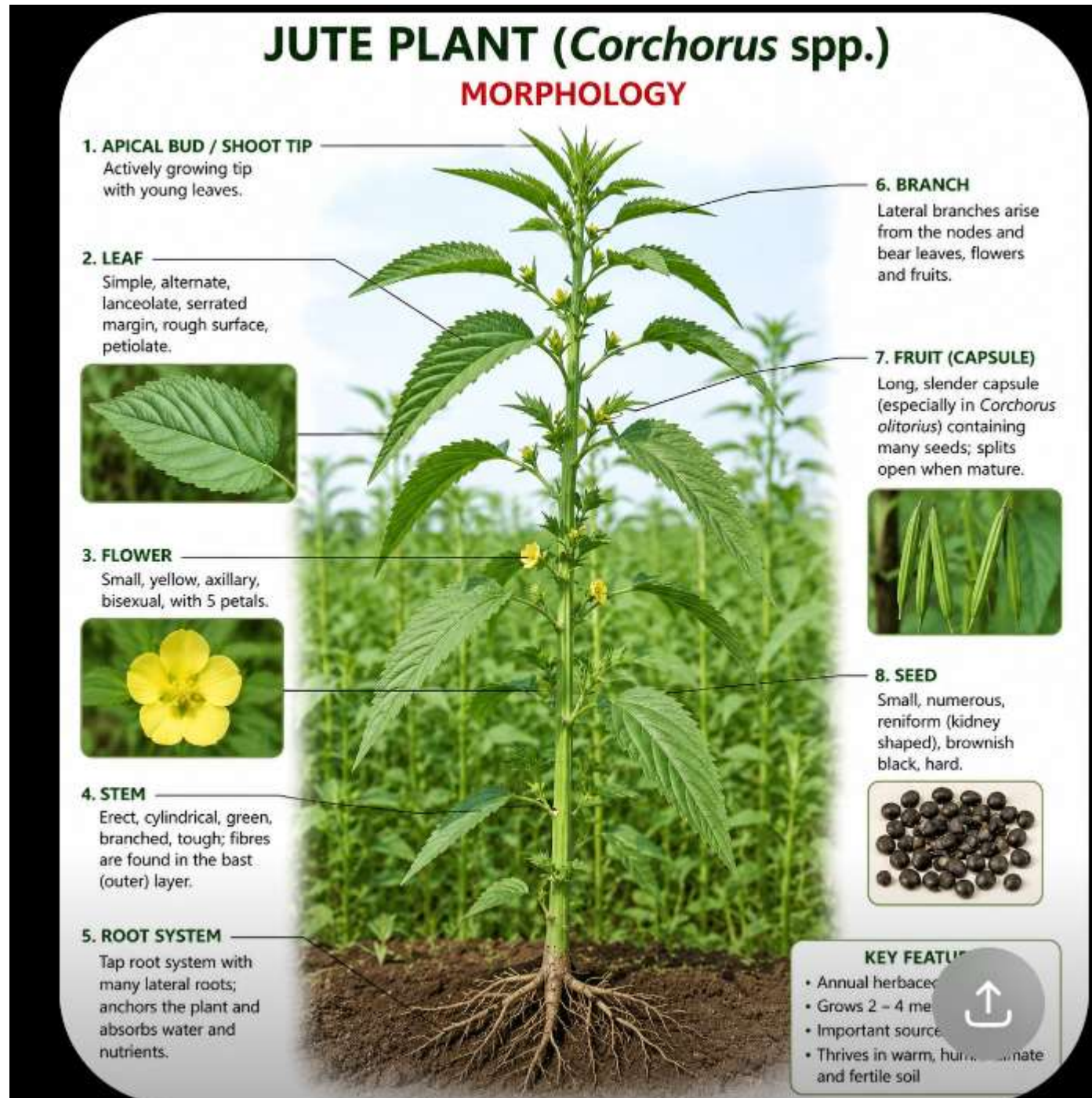
3.0 Botany of Jute

3.1 Taxonomy

Classification Name

Kingdom	Plantae
Division	Angiosperms
Class	Dicotyledonae
Order	Malvales
Family	Malvaceae
Genus	Corchorus
Species	<i>C. capsularis</i> and <i>C. olitorius</i>

Morphology



Root

- Taproot with lateral roots.

Stem

- Tall, erect
- Cylindrical
- Green
- Fibres are found inside the bark

Leaves

- Simple
- Alternate
- Green
- Serrated edges

Flowers

- Small
- Yellow.
- Bisexual.

Fruits

- Capsule (White jute)
- Long cylindrical capsule (Tossa jute)

Seeds

- Small
- Brown or black
- Numerous

Anatomy

Stem contains:

- Epidermis
- Cortex
- Bast fibre (commercial fibre)
- Phloem
- Cambium
- Xylem
- Pith

Structural Forms of Fruits and Seeds**Fruits**

- Dry capsule
- Splits open when mature

Seeds

- Small
- Hard
- Brownish-black
- Good germination if properly stored

3.2 Varieties**White Jute**

Corchorus capsularis

Tossa Jute

Corchorus olitorius

Improved/Recommended Varieties

JRO-524

JRO-632

JRO-8432

JRC-321

JRC-7447

These varieties have:

- Higher fibre yield
- Better disease resistance
- Early maturity

4.0 Production Techniques

Nursery Preparation

Normally, jute is directly sown in the field.

Where seedlings are raised:

Prepare fine seedbed.

Mix compost into the soil.

Ensure proper drainage.

Planting Date

Beginning of rainy season.

Usually April–June in southern Nigeria.

Spacing

20–30 cm × 5–10 cm.

Use of Poly Pots

Rarely used, but when used:

Fill with fertile soil.

Sow 2–3 seeds.

Water regularly.

Transplant after 3–4 weeks.

Nursery Management

Regular watering.

Weeding.

Light shading.

Pest control.

Remove weak seedlings.

Site Selection

Fertile soil.

Good drainage.

Full sunlight.

Easy access.

Land Preparation

Clear vegetation.

Plough.

Harrow.

Make ridges where necessary.

Marking Out and Planting

Mark rows.

Sow seeds shallowly (2–3 cm deep).

Cover lightly.

Cultural Management Practices

Weeding

First: 2–3 weeks after planting.

Second: 5–6 weeks after planting.

Thinning

Remove excess seedlings.

Irrigation

Only during drought.

Pruning

Objectives

Remove diseased branches.

Improve air circulation.

Encourage healthy growth.

Crop Protection

Major Pests

Include: Stem weevils, Aphids, Grasshoppers, and Caterpillars

Control

Good sanitation.

Recommended insecticides.

Crop rotation.

Diseases

Wilt

Anthraxnose

Leaf spot

Root rot

Control

Resistant varieties.

Fungicides.

Proper spacing.

Field sanitation.

Weed Control

Hand weeding.

Hoe weeding.

Mulching.

Herbicides where necessary.

Manure and Fertilizer Application

Apply:

Farmyard manure before planting.

NPK 15:15:15 according to soil test recommendations.

Nitrogen top dressing after establishment.

Harvesting

Harvest 100–120 days after planting when lower leaves begin to shed.

Farm-Level Processing

Cut stems.

Tie into bundles.

Ret in clean water for 10–20 days.

Strip fibres.

Wash fibres.

Dry in the sun.

Bundle for sale.

Grading

Based on:

Fibre length.

Strength.

Colour.

Cleanliness.

Uniformity.

Marketing

Products are sold to:

- Fibre industries.
- Rope manufacturers.
- Bag manufacturers.
- Paper industries.
- Export markets.

Maintenance of Harvesting Implements

- Clean after use.
- Sharpen blades.
- Oil metal parts.
- Store in a dry place.
- Replace damaged parts.

5.0 Production Cycle of Jute

5.1 Life Cycle

Seed selection.

Land preparation.

Planting.

Germination (3–7 days).

Vegetative growth.

Flowering.

Fruit and seed formation.

Harvesting.

Retting.

Fibre extraction.

Drying.

Grading.

Marketing.

Production period: About 100–120 days (3–4 months).

5.2 Yield Capacity and Profitability

Yield Capacity

Fibre yield: 2–3 tonnes per hectare under good management.

Seed yield: 300–500 kg per hectare.

Profitability

Jute is profitable because:

- It has many industrial uses.
- There is increasing demand for biodegradable fibre.
- Production costs are relatively low.
- It provides income from fibre and seeds.
- It creates employment in farming, processing, and marketing.

